

Thrombin-Antithrombin Complex and Fibrinogen Predict Microangiopathic Acute Kidney Injury in Injured Patients

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Rida Zakar (University of Pittsburgh, United States) Gena Topper (Cooper University Health Care, United States) Abiha Abdullah (University of Pittsburgh, United States) Mazen Zenati (University of Pittsburgh, United States) Morgan Mollen (University of Pittsburgh, United States) Tvisha Subhasish (University of Pittsburgh, United States) Reem Younes (University of Pittsburgh, United States) Kimberly Thomas (Vitalant Research Institute, United States) Tanya Egodage (Cooper University Health Care, United States) Bryce Kerlin (Abigail Wexner Research Institute at Nationwide Children's Hospital, United States) Grant Bochicchio (Washington University in Saint Louis, United States) Emily Mihalko (University of Pittsburgh, United States) Julia Coleman (The Ohio State University College of Medicine, United States) Phillip Spinella (Washington University School of Medicine, United States) Susan Shea (University of Pittsburgh, United States) Matthew Neal (University of Pittsburgh Medical Center, United States)

Abstract:

Acute kidney injury (AKI) develops in 20-40% of critically injured patients and contributes to poor outcomes. In sepsis and critical illness, dysregulated coagulation and fibrinolysis are linked to microangiopathic kidney dysfunction, but similar relationships in trauma are not well characterized. We performed a secondary analysis of a single-center randomized controlled trial of severely injured adult trauma patients. Plasma thrombin-antithrombin (TAT), plasmin- α 2-antiplasmin (PAP), and fibrinogen were measured within 2 hours of injury and 6 hours later. The primary outcome was AKI, defined by the RIFLE (Risk, Injury, Failure, Loss, End-stage kidney disease) criteria. Adjusted logistic regression models evaluated biomarker associations with AKI. Receiver operating characteristic analyses identified discriminatory thresholds. Among 149 patients, 30 (20.1%) developed AKI. Patients with AKI had higher TAT and PAP concentrations and lower fibrinogen levels at both time points. The 6-hour TAT-to-fibrinogen ratio was independently associated with AKI (Adjusted OR 7.17; 95% CI 2.34-21.9, $p = 0.001$, AUC = 0.88). Supplemental analyses demonstrated consistent associations for TAT and fibrinogen individually, whereas PAP was not independently predictive. A 6-hour TAT level < 99 pg/mL identified patients unlikely to develop AKI (negative predictive value [NPV] 93.9%). Fibrinogen levels > 173 mg/dL showed a low likelihood of AKI (NPV 88.2%). Elevated TAT and reduced fibrinogen measured 6 hours after injury were independently associated with AKI. This suggests that early coagulation imbalance may reflect evolving microvascular kidney dysfunction in injured patients and potentially warrants the evaluation of the TAT-to-fibrinogen ratio as an early marker of trauma-related AKI. (NCT02535949)

Conflict of interest: COI declared - see note

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Clinical trial registration information (if any): NCT02535949

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Running title: Markers of Microangiopathy in Traumatic AKI

Authors: Rida Zakar¹, Gena V. Topper², Abiha Abdullah¹, Morgan Mollen¹, Tvisha Subhasish¹, Reem Younes¹, Mazen S. Zenati¹, Kimberly A. Thomas^{3,4}, Tanya Egodage², Bryce A. Kerlin^{5,6}, Grant V. Bochicchio⁷, Emily Mihalko¹, Julia R. Coleman⁸, Philip C. Spinella¹, Susan M. Shea^{1,9, ++}, Matthew D. Neal^{1*,++}

Affiliations:

¹Trauma and Transfusion Medicine Research Center, Department of Surgery, University of Pittsburgh, Pittsburgh, PA, USA.

²Department of Surgery, Cooper University Health Care, Camden, NJ, USA.

³Department of Pathology, University of Colorado – Anschutz Medical Campus, Aurora, CO, USA.

⁴Vitalant Research Institute, Denver, CO, USA.

⁵Department of Pediatrics, The Ohio State University College of Medicine, Columbus, OH, USA.

⁶Center for Clinical & Translational Research, Abigail Wexner Research Institute at Nationwide Children's Hospital, Columbus, OH, USA

⁷Department of Surgery, Washington University, St. Louis, MO, USA.

⁸Department of Surgery, The Ohio State University College of Medicine, Columbus, OH, USA.

⁹Department of Bioengineering, University of Pittsburgh, PA, USA.

23 *Corresponding author

24 ** Drs. Neal and Shea served as co-senior authors

25 **Corresponding author:**

26 Matthew D. Neal

27 200 Lothrop St., F1271.2 PUH

28 Pittsburgh, PA 15213

29 Phone: 1- 412-864-5532

30 E-mail: nealm2@upmc.edu

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39 De-identified individual participant data from the TAMPITI trial are available from

40 Washington University upon reasonable request and completion of a data use

41 agreement, subject to institutional and regulatory requirements. The data that support

42 the findings of this study include clinical and biomarker variables used in the present

43 secondary analysis. Requests for access should be directed to the corresponding

44 author. Derived analytic datasets and statistical code generated for this study are

45 available from the corresponding author upon reasonable request.

The parent study protocol and related trial details have been previously published and are publicly available (ClinicalTrials.gov identifier: NCT02535949).

Key Points

- Trauma-associated AKI is multifactorial and may arise in part from microvascular thrombosis driven by early hemostatic dysregulation.
- Increased thrombin generation and reduced fibrinogen are linked to AKI, and their ratio may serve as a marker of organ failure in trauma.

Abstract

Acute kidney injury (AKI) develops in 20–40% of critically injured patients and contributes to poor outcomes. In sepsis and critical illness, dysregulated coagulation and fibrinolysis are linked to microangiopathic kidney dysfunction, but similar relationships in trauma are not well characterized. We performed a secondary analysis of a single-center randomized controlled trial of severely injured adult trauma patients. Plasma thrombin-antithrombin (TAT), plasmin- α 2-antiplasmin (PAP), and fibrinogen were measured within 2 hours of injury and 6 hours later. The primary outcome was AKI, defined by the RIFLE (Risk, Injury, Failure, Loss, End-stage kidney disease) criteria. Adjusted logistic regression models evaluated biomarker associations with AKI. Receiver operating characteristic analyses identified discriminatory thresholds. Among 149 patients, 30 (20.1%) developed AKI. Patients with AKI had higher TAT and PAP concentrations and lower fibrinogen levels at both time points. The 6-hour TAT-to-fibrinogen ratio was independently associated with AKI (Adjusted OR 7.17; 95% CI 2.34-21.9, $p = 0.001$, AUC = 0.88). Supplemental analyses demonstrated consistent associations for TAT and fibrinogen individually, whereas PAP was not independently

predictive. A 6-hour TAT level < 99 pg/mL identified patients unlikely to develop AKI (negative predictive value [NPV] 93.9%). Fibrinogen levels > 173 mg/dL showed a low likelihood of AKI (NPV 88.2%). Elevated TAT and reduced fibrinogen measured 6 hours after injury were independently associated with AKI. This suggests that early coagulation imbalance may reflect evolving microvascular kidney dysfunction in injured patients and potentially warrants the evaluation of the TAT-to-fibrinogen ratio as an early marker of trauma-related AKI. (NCT02535949)

Introduction

Acute kidney injury (AKI) affects a substantial proportion of critically injured trauma patients, with reported incidences ranging from 20% to 45% among those requiring intensive care^{1,2}. AKI occurrence is frequently associated with prolonged ICU stays, increased risk of multi-organ dysfunction, and higher mortality²⁻⁴. Despite its clinical significance, early identification of patients at risk for trauma-associated AKI remains a persistent challenge. Current diagnostic tools, based on serum creatinine and urine output, lack temporal sensitivity and pathophysiologic specificity. These functional markers are influenced by age, sex, muscle mass, and volume status, and they often rise only after renal injury has occurred⁵. As a result, they fail to reliably detect early or subclinical AKI, delaying opportunities for intervention.

The pathophysiology of trauma-related AKI is multifactorial, but mounting evidence implicates endothelial injury and coagulopathy as central drivers of organ dysfunction. Trauma induces a dynamic disturbance in hemostasis, characterized by thrombin generation, fibrinogen consumption, and impaired fibrinolysis, that progresses from

early hypocoagulability to a prothrombotic state within 24-48 hours, impairing microvascular perfusion and promoting tissue ischemia⁶. In the renal circulation, this manifests as fibrin- and platelet-rich microthrombi that are difficult to detect and lack validated circulating markers⁷. Although trauma is not formally classified as a thrombotic microangiopathy⁸, it shares key pathophysiologic features with syndromes such as sepsis-associated disseminated intravascular coagulation (DIC) and atypical hemolytic uremic syndrome (aHUS), including endothelial injury, excessive thrombin generation, and obstruction of the microcirculation⁹⁻¹². This trauma-specific microangiopathic phenotype mirrors the endothelial and coagulation abnormalities seen in sepsis-associated AKI, by driving the formation of microthrombi that obstruct the renal microcirculation^{11,13}. It remains an underrecognized cause of organ dysfunction and highlights the need for biomarkers that can detect early renal microvascular injury *in vivo*. Currently, there is no widely accepted biomarker that predicts microangiopathic AKI, limiting early identification and targeted intervention in trauma-related kidney injury. Given the profound disruptions in hemostasis that occur following traumatic injury, a wide range of biomarkers have been used to characterize and investigate trauma-induced coagulopathy and its downstream effects. Commonly studied markers include platelet count and D-dimer, which provide insight into platelet consumption and fibrin degradation, respectively¹⁴. Other biomarkers, such as prothrombin fragment 1+2 (PF1+2), have been investigated as measures of coagulation activation¹⁵. In the present study, we focused on thrombin-antithrombin (TAT) complex, plasmin- α 2-antiplasmin (PAP) complex, and fibrinogen as they reflect complementary aspects of post-traumatic hemostatic dysfunction^{16,17}. Notably, TAT and PAP have previously investigated in

conditions characterized by widespread microvascular thrombosis⁷, making them possible informative candidates for evaluating the contribution of microangiopathic processes to trauma-associated AKI.

We hypothesized that coagulation biomarkers known to be associated with microvascular thrombosis and their composite ratios would help identify trauma patients at risk for AKI and reflect evolving microvascular dysfunction. To test this, we conducted a secondary analysis of a randomized clinical trial studying the effects of tranexamic acid in severely injured patients which included early and frequent biosampling. We evaluated whether early biomarkers measured within 2 and/or 6 hours of injury could predict AKI following trauma.

Methods

Study design

This was a secondary analysis of the Tranexamic Acid Mechanisms and Pharmacokinetics in Traumatic Injury (TAMPITI) trial (NCT02535949), a randomized, double-blind, placebo-controlled study conducted at an academic Level I trauma center^{18–20}. The trial enrolled 149 adult trauma patients who were randomized to receive either 2g or 4g of intravenous tranexamic acid (TXA) or placebo. The parent study characterized this population through immunologic, coagulation, and pharmacokinetic profiling. For the present analysis, we used biomarker and clinical outcome data prospectively collected during this trial, focusing on arrival and 6-hour timepoints to examine whether early biomarkers could identify patients at risk for trauma-related AKI. The initial sample was drawn on arrival prior to the trial intervention administration,

which occurred within 2 hours of traumatic injury. A second sample was taken 6 hours later, after the intervention was administered.

Data collection

Detailed methodologies regarding trial design, data collection, eligibility criteria, accrual dates, and biomarker processing have been previously published^{18–20}. For the present analysis, we selected TAT, fibrinogen, and PAP *a priori* due to previously reported associations with disorders characterized by consumptive coagulopathy and microvascular thrombosis. In particular, TAT has been studied extensively in DIC, whereas PAP has been investigated in thrombotic microangiopathies such as thrombocytopenic thrombotic purpura (TTP)^{21–23}. In the parent trial, TAT and PAP concentrations were measured by ELISA (Enzygnost TAT, OWMG15, Siemens ; IMUCLONE Alpha-2-Antiplasmin, #603, BioMedica Diagnostics). Fibrinogen was quantified using a mechanical clot-based method (STA-FIBRINOGEN 5, #0064, Diagnostica Stago). Biomarker analyses emphasized the 6-hour post-injury sample to capture hemostatic physiology after initial resuscitation, when trauma-induced coagulopathy continues to evolve⁶, while remaining within the early window in which kidney injury biomarkers such as neutrophil gelatinase-associated lipocalin (NGAL) rise before creatinine²⁴. D-dimer and prothrombin fragment 1+2 (PF1+2) were collected and prospectively measured using an immunoturbidimetric assay (STA LIATEST D-DI, #00515, Diagnostic Stago) and enzyme-linked immunosorbent assay (Prothrombin Fragment 1+2 BioAssay ELISA Kit, #027756, US Biological Life Sciences), respectively.

AKI within 10 days of traumatic injury was defined using the RIFLE (Risk, Injury, Failure, Loss, End-stage kidney disease) classification, based on variations of serum creatinine or urine output over time, and examined as the primary outcome²⁵. Estimated glomerular filtration rate (eGFR) was calculated using the 2021 CKD-EPI creatinine equation²⁶. Secondary outcomes included acute respiratory distress syndrome (ARDS), defined by the Berlin criteria²⁷. Multiple organ failure (MOF) was defined using the Marshall Multiple Organ Dysfunction (MOD) score²⁸. Intensive care unit (ICU)-free days were calculated as 30 minus the number of days in the ICU (range, 0–30 days), and assigned 0 ICU-free days for death within 30 days²⁹. Ventilator-free days were calculated similarly. A limited number of values, from biomarkers of interest and other clinical parameters, fell outside detectable ranges or were missing due to random omissions, as summarized in Supplemental Table S1. These instances were addressed using complete-case analysis.

Statistical analysis and model selection

Statistical analyses were conducted using Stata/SE version 18.5 (StataCorp LLC, College Station, TX). Categorical variables are reported as counts and percentages and compared using Fisher's exact or Chi-square tests, as appropriate. Continuous variables are summarized using medians and interquartile ranges (IQR) and compared using the Wilcoxon rank-sum or Student's t-test based on normality testing with the Shapiro-Wilk test. All tests were two-sided, with a significance threshold of $p < 0.05$. Biomarker quartiles were examined for directional trends with AKI using the nonparametric Cochran-Armitage trend test (Supplementary Table S2). Quartile splitting

was applied to enhance clinical interpretability and to capture linear or threshold effects, consistent with established approaches in trauma research^{30,31}.

Several modeling approaches were undertaken to evaluate associations between hemostatic biomarkers and AKI. These included: (1) direct biomarker modeling, which tested continuous biomarker concentrations; (2) quartile-based modeling, which stratified biomarkers into ordinal groups to assess graded risk; and (3) composite biomarker modeling, which incorporated ratios such as the TAT-to-fibrinogen to ratio. Of these approaches, the composite biomarker model was retained for the main analysis, while results from direct biomarker and quartile-based models are presented in Supplementary Information.

Covariate selection was guided by biological importance, plausibility, clinical context, statistical relevance, and prior literature^{32–34}. Variables nearing significance in univariate analysis ($p < 0.2$) were considered for inclusion. Age and Injury Severity Score (ISS) were forced into all models based on their clinical significance and established role in trauma risk adjustment^{35,36}. Multicollinearity was assessed using pairwise Pearson correlation coefficients prior to each modeling attempt. When variables were clinically dependent or highly correlated ($\rho > 0.6$), one was excluded based on the strength of its association with the outcome in univariate analysis, while also considering overall model parsimony and clinical utility. After variable screening, variables were eliminated using backwards stepwise selection, by eliminating the covariate with the highest p value, refitting the new model, and repeating the process until all covariates were significant (p value cutoff = 0.05) aside from forced-in covariates (age and ISS). Model performance was evaluated using McFadden's pseudo- R^2 , likelihood ratio chi-square, and area

under the curve (AUC). Model calibration was evaluated using the Hosmer-Lemeshow goodness-of-fit test, which compared observed and predicted probabilities of AKI across deciles of risk.

Receiver Operating Characteristic (ROC) analyses were also used to identify optimal thresholds for each biomarker using the highest Youden index (sensitivity + specificity – 1) , with corresponding sensitivity, specificity, positive predictive value (PPV), and negative predictive value (NPV) calculated. Patients were then stratified above and below these thresholds, and outcome comparisons were performed to assess clinical trajectories.

Study approval

This study represents a secondary analysis of data collected as part of the TAMPITI randomized controlled trial. The parent trial was approved by the US Food and Drug Administration (IND 125420), the Institutional Review Board at Washington University School of Medicine (protocol 201412027), and the United States Army Medical Research and Materiel Command Human Research Protections Office. The trial was conducted under an exception from informed consent with community consultation and delayed consent obtained from patients or legally authorized representatives. The current analysis was performed under a Data Use Agreement between Washington University and the University of Pittsburgh (DUA00004631). All procedures adhered to institutional guidelines, Good Clinical Practice, and the Declaration of Helsinki.

Results

Cohort Characteristics

A total of 149 trauma patients were included in the final analysis, of whom 30 (20.1%) developed acute kidney injury as defined by the RIFLE (Risk, Injury, Failure, Loss, End-stage kidney disease) criteria²⁵. Demographic characteristics, including age, sex, and race, were similar between patients with and without AKI (Table 1). In contrast, clinical severity and transfusion exposure differed substantially between groups. Patients with AKI had higher Injury Severity Scores (ISS) (median ISS 34 vs. 18, $p < 0.001$) and were more likely to present with traumatic brain injury (17% vs. 4.2%, $p = 0.015$). Platelet counts on arrival were also lower among patients who developed AKI (Table 1). Patients who developed AKI received substantially higher weight-adjusted transfusion volumes in 24 hours across all blood products, including red blood cells (49.9 vs. 12.9 mL/kg, $p < 0.001$), plasma (23.7 vs. 0.00 mL/kg, $p < 0.001$), and platelets (6.79 vs. 0.00 mL/kg, $p < 0.001$).

Renal function on arrival also differed between groups. As expected, the AKI cohort had a higher serum creatinine (1.37 vs. 1.16 mg/dL, $p < 0.001$) and lower estimated glomerular filtration rate (64.5 vs. 84.0 mL/min/1.73 m², $p < 0.001$) on first measurement after traumatic injury. Hemostatic biomarker levels also differed by AKI status at both measured time points. Patients who developed AKI demonstrated higher concentrations of TAT and PAP complexes on initial sampling and again at 6 hours (all $p < 0.005$; Table 1). In contrast, fibrinogen levels on arrival were lower in patients who developed AKI (158 vs. 199 mg/dL, $p = 0.014$) and at 6 hours (169 vs. 218 mg/dL, $p = 0.010$).

252 Additionally, D-dimer and PF1+2 were evaluated as candidate biomarkers. Neither
253 biomarker demonstrated significant differences between patients who did and did not
254 develop AKI and were therefore not pursued in subsequent modeling analyses.

256 **Biomarker Distributions**

257 Results of nonparametric Cochran-Armitage trend tests demonstrated a consistent,
258 directional relationship between biomarker quartiles and AKI incidence across all
259 analytes (Figure 1, Supplementary Table S2). For TAT complex, a significant upward
260 trend in AKI risk was observed both on arrival ($z = 2.75$, $p = 0.006$) and at 6 hours ($z =$
261 4.76 , $p < 0.001$). Incidence of AKI increased across quartiles from 6.3% in the lowest to
262 35.5% in the highest quartile on arrival, and from 3.3% to 50.0% at 6 hours (Figure
263 1A,B). A similar pattern was observed for PAP complex. At the initial time point, AKI
264 incidence increased from 3.5% to 32.1% across quartiles ($z = 2.24$, $p = 0.025$), while at
265 6 hours, rates rose from 6.9% to 39.3% ($z = 3.07$, $p = 0.002$), suggesting a progressive
266 rise in AKI risk with higher PAP levels (Figure 1C,D).

267 In contrast, fibrinogen demonstrated a significant inverse trend. AKI incidence was
268 highest in the lowest quartile of fibrinogen (30.3%) and decreased as fibrinogen levels
269 increased, falling to 12.5% in the highest quartile ($z = -2.70$, $p = 0.002$ on initial
270 measurement; $z = -0.68$, $p = 0.007$ at 6 hours; Figures 1E,F). This trend is consistent
271 with the expected relationship in which hypofibrinogenemia, reflecting fibrin
272 consumption and deposition within the microvasculature, is associated with higher AKI
273 risk.

Multivariate Models and Risk Gradients

Multivariable models were constructed to account for demographic and clinical covariates that could confound the relationship between biomarkers and AKI. Age and Injury Severity Score (ISS) were forced into all models. Serum creatinine obtained on arrival was included to approximate pre-injury renal function. Additional covariates such as mechanism of injury, sex, shock index, available thromboelastography (TEG) parameters (reaction time, alpha angle, maximal amplitude, and lysis at 30 minutes), tranexamic acid (TXA) dose, 6-hour TXA concentration, and venous thromboembolism chemoprophylaxis status, were considered when they approached significance in univariate analyses ($p < 0.2$). This strategy ensured that observed biomarker associations reflected independent contributions beyond injury severity and initial renal status. Detailed model development is detailed in Supplemental Tables S4–S6. TXA dose was not associated with AKI in univariate and adjusted analyses. 6-hour TXA concentration met inclusion criteria for multivariable modeling but was eliminated during backward stepwise selection, indicating a lack of independent contribution after adjustment. Similarly, chemoprophylaxis status was not associated with AKI incidence in univariate analysis and was excluded from final models (Supplemental Table S3). In a preliminary analysis, both direct and quartile-based models were used to evaluate associations between hemostatic markers and AKI. When modeled as continuous variables, higher 6-hour TAT concentrations were independently associated with increased odds of AKI, whereas higher 6-hour fibrinogen concentrations were independently associated with decreased odds. (Supplementary Table S4). Predictive margins demonstrated a consistent risk gradient, with higher TAT levels linked to

progressively greater predicted probabilities of AKI and higher fibrinogen levels linked to lower predicted probabilities (Supplementary Figure S4). In contrast, initial biomarker concentrations, PAP at either time point, and available TEG parameters did not remain independently predictive after covariate adjustment and were eliminated from final models. Importantly, other coagulation markers, including platelet count on arrival, INR, D-dimer, and prothrombin fragment 1+2, did not demonstrate independent associations with AKI and were not retained in adjusted models. Among six candidate biomarkers evaluated, only TAT and fibrinogen remained independently associated with AKI after multivariable adjustment (Supplemental Tables S4–S6). Additionally, quartile-based modeling showed independent stepwise trends of AKI incidence across quartiles of 6-hour TAT and fibrinogen measurement, with a marked disruption observed after the first quartile of measurement (Supplementary Table and Figure S5).

Building on these findings, a composite biomarker model was developed to capture the balance between thrombin generation and fibrinogen depletion. Of the measures tested, the 6-hour TAT-to-fibrinogen ratio showed the strongest and most consistent association with AKI. In the final multivariable model, a higher ratio was independently associated with increased odds of AKI (aOR: 7.17; 95% CI: 2.34-21.9; $p = 0.001$) (Table 2). The model explained a substantial proportion of outcome variability (McFadden's pseudo- $R^2 = 0.42$) and demonstrated excellent discrimination (AUC = 0.88). Calibration assessed by the Hosmer-Lemeshow test across deciles of risk showed no evidence of lack of fit ($\chi^2(8) = 7.25$, $p = 0.510$). Predictive margins from this model illustrated a clear gradient,

with increasing TAT-to-fibrinogen ratios corresponding to sharply higher predicted probabilities of AKI (Figure 2).

Discrimination and Thresholds

Receiver operating characteristic (ROC) analysis was used to assess the discriminative performance of TAT, fibrinogen, and PAP for predicting AKI on arrival and 6 hours (Figure 3). Among all biomarkers, 6-hour TAT demonstrated the highest area under the curve (AUC) at 0.819 (95% CI, 0.730–0.909), with an optimal threshold of ≥ 99 pg/mL yielding a sensitivity of 85.2% and specificity of 67.4%. This threshold corresponded to a NPV of 93.9%. 6-hour fibrinogen showed moderate discriminative performance (AUC 0.664, 95% CI, 0.541–0.786), with an optimal threshold ≤ 173 mg/dL. This cut-off achieved a specificity of 81.2% but a lower sensitivity of 57.7%, with a PPV of 44.1% and NPV of 88.2%. With this higher specificity, 6-hour fibrinogen demonstrates an enhanced capability in predicting patients who would not develop AKI. 6-hour PAP showed intermediate performance (AUC 0.705, 95% CI, 0.598–0.813), with a cut-off of ≥ 4302 pg/mL resulting in sensitivity and specificity values of 65.4% and 73.0%, respectively.

The discriminative capacity of biomarkers on arrival was consistently lower than their 6-hour counterparts. Detailed diagnostic metrics, including PPV and NPV for each threshold, are presented in Table 3. These thresholds were subsequently used to stratify patients for downstream outcome comparisons.

Stratified Clinical Outcomes

Patients with high TAT (≥ 99 pg/mL) had higher incidence of MOF (51% vs. 17% ; $p < 0.001$) and ARDS (17% vs. 1.5% ; $p = 0.005$). These patients experienced prolonged hospital stays (14 [8.5–21] vs. 8.0 [5.0–11] days ; $p < 0.001$). Mechanical ventilation was more frequent (76% vs. 50% ; $p = 0.005$). In-hospital mortality was numerically higher in the high TAT group (4/53, 7.6% vs. 1/66, 1.5%), but not statistically significant ($p = 0.170$). High PAP levels (≥ 4302 pg/mL) were observed in 41 patients, who demonstrated higher rates of MOF (51% vs. 22% ; $p = 0.001$), ARDS (17% vs. 4.1% ; $p = 0.033$), and mechanical ventilation (85% vs. 47% ; $p < 0.001$). Hospital length of stay was extended (13 [8.5–22] vs. 8.5 [5.0–13] days ; $p = 0.005$), with fewer ICU-free days (26 [14–28] vs. 28 [26–30] ; $p < 0.001$) and ventilator-free days (26 [14–28] vs. 28 [26–30] ; $p < 0.001$). Mortality did not differ significantly (4/41, 9.8% vs. 1/74, 1.4% ; $p = 0.054$). Patients with low fibrinogen (≤ 173 mg/dL) had increased rates of MOF (47% vs. 23% ; $p = 0.007$) and fewer ICU-free days (27 [17–28] vs. 28 [26–30] ; $p = 0.011$). Other outcomes, including ARDS, mechanical ventilation, and mortality, were numerically worse but not statistically significant. All outcome comparisons are summarized in Table 4.

Discussion

In this secondary analysis of a randomized controlled trial^{18–20}, we identified a trauma-related AKI signal characterized by early coagulation imbalance. 6-hour TAT and fibrinogen remained independently associated with AKI after covariate adjustment, and each provided clinically meaningful discrimination. When combined, the 6-hour TAT-to-

fibrinogen ratio yielded the strongest overall performance, supporting its use as the primary composite metric in this framework. The observed AKI incidence (20.1%) was consistent with prior trauma cohorts, acknowledging that reported rates vary by definition and population^{1,2}.

The ratio finding provides a mechanistic frame that is more informative than either biomarker in isolation. TAT serves as a surrogate of *in vivo* thrombin generation, while fibrinogen reflects substrate availability and consumption within early trauma-induced coagulopathy. The TAT-to-fibrinogen ratio therefore captures thrombin excess relative to fibrinogen depletion, combining two dimensions of hemostatic dysregulation into a single metric. This is consistent with the broader rationale for selecting biomarkers validated in syndromes where microvascular thrombosis is prominent, including sepsis-associated AKI and DIC^{21,37}. In those contexts, measurable disturbances in coagulation and fibrinolysis reflect thromboinflammatory microvascular injury, and our data suggest that a similar composite signal may be detectable early after trauma, even if the upstream drivers differ. Notably, neither platelet count, D-dimer, nor PF1+2 demonstrated independent associations with AKI, suggesting that the observed signal may reflect specific aspects of thrombin generation and consumptive coagulopathy rather than generalized coagulation activation alone.

The relationship between trauma-induced coagulopathy (TIC), DIC, and trauma-associated microvascular thrombosis remains incompletely defined³⁸. Although these syndromes share abnormalities in coagulation and fibrinolysis, they are not universally considered synonymous entities, and trauma studies have historically characterized TIC using trauma-specific coagulation abnormalities rather than DIC scoring systems^{39,40}.

Accordingly, our findings are best interpreted within the framework of trauma-associated hemostatic dysregulation and microvascular thrombosis rather than overt DIC alone. The differential performance of initial versus 6-hour measurements likely reflects both timing and physiology. Immediately after injury, biomarker values are susceptible to rapid shifts from hemorrhage, dilution, early transfusion and resuscitation, and acute inflammatory responses. These factors can obscure relationships with downstream organ dysfunction by increasing biologic and treatment-related variability. By 6 hours, the post-resuscitation state may be more stable, improving the signal-to-noise ratio and allowing hemostatic patterns that track with evolving organ injury to emerge more clearly. This interpretation is also consistent with the observation that fibrinogen trajectories may diverge across patients: in some, fibrinogen rises as part of an acute phase response, whereas in others, it may fail to rise because consumption outpaces synthesis or circulating levels are diluted. That divergence plausibly separates a more compensated response from a pattern of persistent coagulopathy and microvascular vulnerability that tracks with AKI.

From a translational standpoint, the derived thresholds provide an interpretable approach to early risk stratification, while also highlighting that cutoffs are context- and time-dependent. Importantly, the proposed TAT-to-fibrinogen ratio should be considered hypothesis-generating and requires validation in independent trauma cohorts before adoption as a risk-stratification tool. In our cohort, a 6-hour TAT threshold of 99 pg/mL identified patients unlikely to develop AKI (NPV 93.9%), and fibrinogen levels above 173 mg/dL similarly marked low risk (NPV 88.2%). Prior studies in other critical illness settings support these findings, but with different absolute thresholds that likely reflect

population characteristics, sampling windows, and disease trajectory. Minami et al. reported postoperative TAT levels independently associated with AKI after pediatric cardiopulmonary bypass, with a threshold of 61.2 ng/mL⁴¹, a setting where contact activation contributes to excessive thrombin generation⁴². In ICU cohorts, Xie et al. demonstrated strong discrimination for AKI nondevelopment using an antithrombin-to-creatinine ratio⁴³, and Zhang et al. observed that fibrinogen and antithrombin activity differed across AKI stages, with more advanced stages in patients with fibrinogen < 150 mg/dL or antithrombin activity < 80%²². Although antithrombin activity was not available in our dataset, the inverse functional relationship between thrombin generation (captured by TAT) and antithrombin supports a convergent thrombin-driven axis across settings. Collectively, these comparisons highlight that while thromboinflammatory signatures recur, thresholds must be calibrated to syndrome, age, and timing. Furthermore, patients who developed AKI received substantially greater transfusion volumes than those who did not develop AKI. In trauma populations, transfusion burden has previously been associated with subsequent organ dysfunction, including AKI, and packed red blood cell (pRBC) exposure has been linked to inflammatory activation and endothelial injury^{44–46}. While the present study was not designed to isolate the effects of individual blood products, we observed a consistent signal across red blood cells, plasma and platelet transfusion. For this reason, we interpreted weight-adjusted transfusion volume as a surrogate of overall transfusion burden rather than attributing the observed associations to any single product. Larger trauma studies have explored the independent relationship between transfusion exposure and adverse outcomes in greater detail⁴⁵.

This study has important limitations. As a single-center secondary analysis with modest sample size, model complexity was constrained and residual confounding remains possible. The cohort was composed of predominantly young, Black or African male patients with low baseline comorbidity burden, which may help isolate trauma-related mechanisms but limits generalizability and reduces the ability to interrogate sex- and hormone-related modifiers. We did not evaluate oral contraception or hormone therapy. In addition, because the parent trial was not designed as an AKI study, creatinine was obtained through clinical workflows rather than standardized to the 6-hour research draw, precluding precise temporal alignment of renal function with the biomarker time point. While adjustment included injury severity and other clinical covariates, physiologic changes linked to shock and TBI may not be fully captured by available variables. This analysis does not disentangle the overlapping contributions of coagulation, fibrinolysis, inflammation, and endotheliopathy, and it lacks histologic confirmation of renal microvascular thrombosis, although animal models support this mechanism^{9,10}. Accordingly, these findings should be interpreted as associative and hypothesis-generating.

The intent of this work is early identification rather than definitive causal inference, aligning with the broader biomarker-based AKI literature that uses early signals to enable timely nephroprotective measures and structured supportive bundles⁴⁷. Future studies should validate these findings in larger, multicenter trauma cohorts with greater demographic and clinical heterogeneity, and incorporate established kidney injury biomarkers such as Cystatin C, NGAL and liver-type fatty acid binding protein (L-FABP) to improve prediction and clarify temporal sequencing between hemostatic

activation and renal dysfunction^{48–50}. Mechanistic work evaluating sex- and hormone-related differences in endothelial response and microvascular thrombosis will be important⁵¹. Although TXA was not associated with AKI in this analysis, its potential impact on microvascular thrombosis and fibrinolytic pathways remains relevant for future investigation, particularly in relation PAP.

Overall, our data support the concept that early thrombin generation relative to fibrinogen availability identifies a trauma-related AKI risk phenotype, and warrant continued evaluation of TAT, fibrinogen, and their ratio as clinically interpretable tools for early risk stratification. These findings require validation in independent trauma cohorts or a future multicenter study, particularly given the predominantly young, African American, single-center population studied here, before broader clinical application.

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Conflicts of interests: M.D.N. serves as the Chief Medical Officer for Haima Therapeutics and has received research funding from the NIH, DoD, Takeda and Haemonetics. He serves on the advisory board and receives honoraria from Haemonetics, Takeda, Octapharma, and CellPhire. P.C.S. is the co-founder and Chief Medical Officer of Kalocyte. He has received consultant fees from Cerus Corporation, and serves on the advisory board of Octapharma, Haima Therapeutics, and Grifols. The remaining authors have nothing to disclose.

Contributors

R.Z. performed the data analysis and drafted the manuscript. M.S.Z. and E.M. provided analytical guidance and oversight. A.A., M.M., T.S. and R.Y. contributed to data interpretation and manuscript editing. K.A.T., S.M.S., P.C.S., and G.V.B. contributed to the design and execution of the parent TAMPITI trial. T.E., B.K., J.R.C. provided collaborative input and interpretation. M.D.N. and S.M.S. supervised the study, provided financial support, reviewed data and data interpretation, and edited the manuscript. All authors reviewed and approved the final manuscript.

References

1. Haines RW, Fowler AJ, Kirwan CJ, Prowle JR. The incidence and associations of acute kidney injury in trauma patients admitted to critical care: A systematic review and meta-analysis. *J Trauma Acute Care Surg*. 2019;86(1):141-147. doi:10.1097/ta.0000000000002085
2. Hatton GE, Harvin JA, Wade CE, Kao LS. Importance of duration of acute kidney injury after severe trauma: a cohort study. *Trauma Surg Acute Care Open*. 2021;6(1):e000689. doi:10.1136/tsaco-2021-000689

3. Ahmed N, Mathew RO, Kuo Y, Asif A. Risk of in-hospital mortality in severe acute kidney injury after traumatic injuries: a national trauma quality program study. *Trauma Surg Acute Care Open*. 2021;6(1):e000635. doi:10.1136/tsaco-2020-000635
4. Büchele G, Rehm M, Halbgebauer R, Rothenbacher D, Huber-Lang M. Trauma-related acute kidney injury during inpatient care of femoral fractures increases the risk of mortality: A claims data analysis. *Am J Med Open*. 2022;8:100009. doi:10.1016/j.ajmo.2022.100009
5. Villegas CV, Gorman E, Liu FM, Winchell RJ. Acute kidney injury in the acute care surgery patient: What you need to know. *J Trauma Acute Care Surg*. 2024;97(6):839-848. doi:10.1097/ta.0000000000004401
6. Moore EE, Moore HB, Kornblith LZ, et al. Trauma-induced coagulopathy. *Nat Rev Dis Primer*. 2021;7(1). doi:10.1038/s41572-021-00264-3
7. Zakar R, Neal MD, Shea SM. Targeting hemostatic enzymes: from mechanistic insights to therapeutic frontiers. *Curr Opin Hematol*. Published online 15 July 2025. doi:10.1097/moh.0000000000000884
8. George JN, Nester CM. Syndromes of Thrombotic Microangiopathy. *N Engl J Med*. 2014;371(7):654-666. doi:10.1056/nejmra1312353
9. Plautz WE, Haldeman SH, Dyer MR, et al. Reduced cleavage of von willebrand factor by ADAMTS13 is associated with microangiopathic acute kidney injury following trauma. *Blood Coagul Fibrinolysis*. 2022;33(1):14-24. doi:10.1097/MBC.0000000000001089
10. Kleinveld DJB, Simons DDG, Dekimpe C, et al. Plasma and rhADAMTS13 reduce trauma-induced organ failure by restoring the ADAMTS13-VWF axis. *Blood Adv*. 2021;5(17):3478-3491. doi:10.1182/bloodadvances.2021004404
11. Iba T, Helms J, Maier CL, Levi M, Scarlatescu E, Levy JH. The role of thromboinflammation in acute kidney injury among patients with septic coagulopathy. *J Thromb Haemost*. 2024;22(6):1530-1540. doi:10.1016/j.jtha.2024.02.006
12. Chang R, Cardenas JC, Wade CE, Holcomb JB. Advances in the understanding of trauma-induced coagulopathy. *Blood*. 2016;128(8):1043-1049. doi:10.1182/blood-2016-01-636423
13. Schult L, Halbgebauer R, Karasu E, Huber-Lang M. Glomerular injury after trauma, burn, and sepsis. *J Nephrol*. 2023;36(9):2417-2429. doi:10.1007/s40620-023-01718-5
14. Kornblith LZ, Moore HB, Cohen MJ. Trauma-Induced Coagulopathy: The Past, Present, and Future. *J Thromb Haemost JTH*. 2019;17(6):852-862. doi:10.1111/jth.14450

15. Meledeo MA, Peltier G, McIntosh C, Mucha N, Bynum J, Cap AP. Trauma Biomarkers in Plasma during the First 24 Hours. *Blood*. 2021;138:591. doi:10.1182/blood-2021-147498
16. Gando S, Shiraishi A, Wada T, et al. A multicenter prospective validation study on disseminated intravascular coagulation in trauma-induced coagulopathy. *J Thromb Haemost JTH*. 2020;18(9):2232-2244. doi:10.1111/jth.14931
17. Hayakawa M, Sawamura A, Gando S, et al. Disseminated intravascular coagulation at an early phase of trauma is associated with consumption coagulopathy and excessive fibrinolysis both by plasmin and neutrophil elastase. *Surgery*. 2011;149(2):221-230. doi:10.1016/j.surg.2010.06.010
18. Spinella PC, Thomas KA, Turnbull IR, et al. The Immunologic Effect of Early Intravenous Two and Four Gram Bolus Dosing of Tranexamic Acid Compared to Placebo in Patients With Severe Traumatic Bleeding (TAMPITI): A Randomized, Double-Blind, Placebo-Controlled, Single-Center Trial. *Front Immunol*. 2020;11:2085. doi:10.3389/fimmu.2020.02085
19. Spinella PC, Bochicchio K, Thomas KA, et al. The risk of thromboembolic events with early intravenous 2- and 4-g bolus dosing of tranexamic acid compared to placebo in patients with severe traumatic bleeding: A secondary analysis of a randomized, double-blind, placebo-controlled, single-center trial. *Transfusion (Paris)*. 2022;62(S1). doi:10.1111/trf.16962
20. Stitt G, Spinella PC, Bochicchio GV, Roberts I, Downes KJ, Zuppa AF. Population pharmacokinetic modelling and simulation of tranexamic acid in adult trauma patients. *Br J Clin Pharmacol*. 2024;90(8):1932-1941. doi:10.1111/bcp.16075
21. Bai H, Shen L, Zhang H, Tang N. Clinical value of TAT, PIC and t-PAIC as predictive markers for severe sepsis in pediatric patients. *Front Pediatr*. 2024;12:1336583. doi:10.3389/fped.2024.1336583
22. Zhang G, Zhang L, Si S, et al. Fibrinogen and antithrombin III are associated with in-hospital mortality among critically ill patients with acute kidney injury. *Ren Fail*. 2022;44(1):1948-1957. doi:10.1080/0886022x.2022.2142138
23. El Otmani H, Frunt R, Smits S, et al. Plasmin-cleaved von Willebrand factor as a biomarker for microvascular thrombosis. *Blood*. 2024;143(20):2089-2098. doi:10.1182/blood.2023021265
24. Dent CL, Ma Q, Dastrala S, et al. Plasma neutrophil gelatinase-associated lipocalin predicts acute kidney injury, morbidity and mortality after pediatric cardiac surgery: a prospective uncontrolled cohort study. *Crit Care*. 2007;11(6):R127. doi:10.1186/cc6192
25. Bellomo R, Ronco C, Kellum JA, Mehta RL, Palevsky P, the ADQI workgroup. Acute renal failure – definition, outcome measures, animal models, fluid therapy and

information technology needs: the Second International Consensus Conference of the Acute Dialysis Quality Initiative (ADQI) Group. *Crit Care*. 2004;8(4). doi:10.1186/cc2872

26. Inker LA, Eneanya ND, Coresh J, et al. New Creatinine- and Cystatin C–Based Equations to Estimate GFR without Race. *N Engl J Med*. 2021;385(19):1737-1749. doi:10.1056/nejmoa2102953
27. Acute Respiratory Distress Syndrome: The Berlin Definition. *JAMA*. 2012;307(23). doi:10.1001/jama.2012.5669
28. Marshall JC. THE MULTIPLE ORGAN DYSFUNCTION (MOD) SCORE: 685. *Shock*. 1997;7(Supplement):172. doi:10.1097/00024382-199703001-00698
29. Cho H, Wendelberger B, Gausche-Hill M, et al. ICU-free days as a more sensitive primary outcome for clinical trials in critically ill pediatric patients. *JACEP Open*. 2021;2(4):e12479. doi:10.1002/emp2.12479
30. Sloos PH, Dujardin RWG, Meijers JCM, et al. The role of factor V in trauma-induced coagulopathy: an observational and experimental study. *Res Pract Thromb Haemost*. 2025;9(4):102857. doi:10.1016/j.rpth.2025.102857
31. Zhang Y, Yan Y, Sun L, Wang Y. Stress hyperglycemia ratio is a risk factor for mortality in trauma and surgical intensive care patients: a retrospective cohort study from the MIMIC-IV. *Eur J Med Res*. 2024;29(1):558. doi:10.1186/s40001-024-02160-4
32. Tran TT, Yun G, Kim S. Artificial intelligence and predictive models for early detection of acute kidney injury: transforming clinical practice. *BMC Nephrol*. 2024;25(1):353. doi:10.1186/s12882-024-03793-7
33. Cartin-Ceba R, Kashiouris M, Plataki M, Kor DJ, Gajic O, Casey ET. Risk Factors for Development of Acute Kidney Injury in Critically Ill Patients: A Systematic Review and Meta-Analysis of Observational Studies. *Crit Care Res Pract*. 2012;2012:1-15. doi:10.1155/2012/691013
34. Bursac Z, Gauss CH, Williams DK, Hosmer DW. Purposeful selection of variables in logistic regression. *Source Code Biol Med*. 2008;3(1):17. doi:10.1186/1751-0473-3-17
35. Shea SM, Mihalko EP, Lu L, et al. Doing more with less: low-titer group O whole blood resulted in less total transfusions and an independent association with survival in adults with severe traumatic hemorrhage. *J Thromb Haemost*. 2024;22(1):140-151. doi:10.1016/j.jtha.2023.09.025
36. Haider AH, Hashmi ZG, Zafar SN, et al. Developing best practices to study trauma outcomes in large databases: An evidence-based approach to determine the best

mortality risk adjustment model. *J Trauma Acute Care Surg.* 2014;76(4):1061-1069.
doi:10.1097/ta.0000000000000182

37. Fang X, Fu W, Xu L, Qiu Y. Analysis of the diagnostic value of coagulation markers and coagulation function indices on the occurrence of DIC in sepsis and its prognosis. *Allergol Immunopathol (Madr).* 2024;52(5):65-72.
doi:10.15586/aei.v52i5.1119

38. Iba T, Helms J, Neal MD, Levy JH. Mechanisms and management of the coagulopathy of trauma and sepsis: trauma-induced coagulopathy, sepsis-induced coagulopathy, and disseminated intravascular coagulation. *J Thromb Haemost.* 2023;21(12):3360-3370. doi:10.1016/j.jtha.2023.05.028

39. Johansson PI, Sørensen AM, Perner A, et al. Disseminated intravascular coagulation or acute coagulopathy of trauma shock early after trauma? An observational study. *Crit Care.* 2011;15(6):R272. doi:10.1186/cc10553

40. Gando S, Wada H, Thachil J, Scientific and Standardization Committee on DIC of the International Society on Thrombosis and Haemostasis (ISTH). Differentiating disseminated intravascular coagulation (DIC) with the fibrinolytic phenotype from coagulopathy of trauma and acute coagulopathy of trauma-shock (COT/ACOTS). *J Thromb Haemost JTH.* 2013;11(5):826-835. doi:10.1111/jth.12190

41. Minami S, Nagafuchi H, Yamada K, Abe T, Takeuchi I. Association Between Thrombin–Antithrombin Complex and Acute Kidney Injury After Pediatric Cardiopulmonary Bypass Surgery: A Single-Center Retrospective Observational Study. *Clin Appl Thromb.* 2023;29:10760296231184465.
doi:10.1177/10760296231184465

42. Levy JH, Alexander PMA, Wolberg AS, et al. ECMO-induced coagulopathy: strategic initiatives for research and clinical practice (a workshop report of the NHLBI). *Blood Vessels Thromb Hemost.* 2025;2(2):100064. doi:10.1016/j.bvth.2025.100064

43. Xie Y, Tian R, Jin W, et al. Antithrombin III expression predicts acute kidney injury in elderly patients with sepsis. *Exp Ther Med.* Published online 9 December 2019.
doi:10.3892/etm.2019.8305

44. Perkins ZB, Captur G, Bird R, Gleeson L, Singer B, O'Brien B. Trauma induced acute kidney injury. Burdmann EA, ed. *PLOS ONE.* 2019;14(1):e0211001.
doi:10.1371/journal.pone.0211001

45. Risa EL, Loudon AM, Pawar OS, Moorman ML, Rushing AP, Ross JT. High fresh frozen plasma to packed red blood cell ratios are associated with lower rates of acute kidney injury and acute respiratory distress syndrome in Trauma Quality Improvement Program: A propensity score-matched analysis. *J Trauma Acute Care Surg.* 2026;100(5):760-767. doi:10.1097/TA.0000000000004899

46. Urner M, Herrmann IK, Buddeberg F, et al. Effects of blood products on inflammatory response in endothelial cells in vitro. *PloS One*. 2012;7(3):e33403. doi:10.1371/journal.pone.0033403
47. Strauß C, Booke H, Forni L, Zarbock A. Biomarkers of acute kidney injury: From discovery to the future of clinical practice. *J Clin Anesth*. 2024;95:111458. doi:10.1016/j.jclinane.2024.111458
48. Klementa V, Petejova N, Zadrazil J, et al. Prediction of Akute Kidney Injury Development in Critically Ill Septic Patients Based on NGAL Determination. *Physiol Res*. Published online 31 December 2024:1001-1011. doi:10.33549/physiolres.935336
49. Wilnes B, Castello-Branco B, Branco BC, Sanglard A. Urinary L-FABP as an Early Biomarker for Pediatric Acute Kidney Injury Following Cardiac Surgery with Cardiopulmonary Bypass: A Systematic Review and Meta-Analysis. *Int J Mol Sci*. Published online 2024.
50. Nguyen Hai G, Gia BN, Thanh HD, et al. The Urinary Liver-Type Fatty Acid Binding Protein (L-FABP) in Early Detection and Outcome Prediction of Sepsis-Associated Acute Kidney Injury. *Arch Acad Emerg Med*. 2025;13(1):e26. doi:10.22037/aaemj.v13i1.2525
51. Coleman JR, Gumina R, Hund T, et al. Sex Dimorphisms in Coagulation: Implications in Trauma-induced Coagulopathy and Trauma Resuscitation. *Am J Hematol*. 2024;99(Suppl 1):S28-S35. doi:10.1002/ajh.27296

Figure legends

Figure 1. AKI incidence by biomarker quartiles on arrival and at 6 hours.

Bar charts depict the number and proportion of acute kidney injury (AKI) events across increasing quartiles (Q1–Q4) of thrombin-antithrombin complex (TAT), plasmin- α 2-antiplasmin complex (PAP), and fibrinogen measured on arrival and 6 hours later. Significant upward trends in AKI incidence were observed for TAT and PAP, and significant downward trends for fibrinogen. Trends were assessed using the nonparametric Cochran-Armitage test. Panel A: Arrival TAT. Panel B: Six-hour TAT. Panel C: Arrival PAP. Panel D: Six-hour PAP. Panel E: Arrival fibrinogen. Panel F: Six-hour fibrinogen.

Figure 2. Predicted probability of acute kidney injury (AKI) derived from the composite biomarker model. Marginal predicted probabilities of AKI are plotted across levels of 6 hour TAT-to-fibrinogen ratio. The model controlled for age, injury severity score, and creatinine obtained on arrival. Shaded regions indicate 95% confidence intervals. Increasing TAT-to-fibrinogen ratios show rising AKI probabilities.

Abbreviations: AKI, acute kidney injury ; TAT, thrombin-antithrombin complex.

Figure 3. ROC curves for early and late biomarkers predicting AKI. Receiver operating characteristic (ROC) curves comparing predictive performance of TAT, PAP, and fibrinogen on arrival and at 6 hours for AKI.

Abbreviations: AKI, acute kidney injury ; TAT, thrombin-antithrombin complex ; PAP, plasmin- α 2-antiplasmin complex.

Tables

Table 1. Baseline demographic, clinical, and laboratory characteristics stratified by acute kidney injury (AKI) status.

Characteristic	No AKI (N = 119)	AKI (N = 30)	p-value
Patient Demographics			
Age, (yr), median [IQR]	27 [22–37]	34 [24–51]	0.087
Male sex, <i>n</i> (%)	102 (85.7%)	29 (97%)	0.100
BMI, kg/m ² , median [IQR]	25.5 [23.4–29.3]	26.3 [22.9–30.6]	0.576
Race, <i>n</i> (%)			1.000
Black or African American	102 (85.7%)	27 (90%)	
White or Caucasian	15.0 (12.6%)	3.0 (10%)	
Other or Unknown	2.00 (1.70%)	0.0 (0.0%)	
Injury Characteristics			
Injury Severity Score (ISS), median [IQR]	18 [13–29]	34 [22–41]	< 0.001
Mechanism of injury, <i>n</i> (%)			0.165
Penetrating	97 (82%)	21 (70%)	
Blunt	22 (19%)	9.0 (30%)	
Admission SBP, mmHg, median [IQR]	119 [98–140]	113 [89–139]	0.281
Admission HR, beats per minute, median [IQR]	99 [84–119]	108 [85–128]	0.464
Shock Index, median [IQR]	0.85 [0.66–1.10]	0.95 [0.71–1.34]	0.204
Confirmed traumatic brain injury, <i>n</i> (%)	5.0 (4.2%)	5.0 (17%)	0.015
GCS ≤ 8, <i>n</i> (%)	20 (16.8%)	9.0 (30%)	0.103
Treatments and Transfusion Volumes			
Given TXA dose, <i>n</i> (%)			0.720
Placebo	39 (33%)	11 (34%)	
2g	41 (35%)	8.0 (27%)	
4g	39 (33%)	11 (37%)	
Total weight-adjusted transfusion volume (24 hours), mL/kg, median [IQR]	23.2 [9.43–60.2]	81.8 [31.8–219]	< 0.001
RBCs, mL/kg, median [IQR]	12.9 [4.42–31.8]	49.9 [20.9–111]	< 0.001
Platelets, mL/kg, median [IQR]	0.00 [0.00–2.58]	6.79 [0.00–15.2]	< 0.001
Plasma, mL/kg, median [IQR]	0.00 [0.00–15.7]	23.7 [2.00–75.2]	< 0.001
Cryoprecipitate, mL/kg, median [IQR]	0.00 [0.00–0.00]	0.00 [0.00–3.84]	< 0.001
Laboratory Values on Arrival			
Creatinine, mg/dL, median [IQR]	1.16 [0.97–1.33]	1.37 [1.19–1.60]	< 0.001
eGFR, mL/min/1.73 m ² , median [IQR]	84.0 [72.0–100]	64.5 [54.5–85.5]	< 0.001
Hb, g/dL, median [IQR]	12.4 [10.6–13.5]	12.3 [10.7–13.3]	0.756
Lactate, mmol/L, median [IQR]	4.5 [3.3–8.3]	6.6 [2.8–9.1]	0.397
INR, median [IQR]	1.12 [1.06–1.22]	1.14 [1.05–1.27]	0.637
Platelet count, x10 ⁹ /L, median [IQR]	231 [185–282]	184 [143–238]	0.008
Biomarker Concentrations on Arrival			
TAT, pg/mL, median [IQR]	126 [53.0–225]	195 [131–328]	0.007
Fibrinogen, mg/dL, median [IQR]	199 [152–247]	158 [132–188]	0.014
PAP, pg/mL, median [IQR]	2058 [947.0–5736]	3762 [2190–8543]	0.039
D-dimer, µg/mL, median [IQR]	3.325 [1.960–9.092]	6.113 [2.898–14.47]	0.131
Prothrombin fragment 1+2, pg/mL, median [IQR]	3895 [2778–5284]	3111 [1715–5731]	0.220
Biomarker Concentrations at Six Hours			
TAT, pg/mL, median [IQR]	63.5 [36.5–126]	163 [108–387]	< 0.001
Fibrinogen, mg/dL, median [IQR]	218 [189–276]	169 [146–239]	0.010
PAP, pg/mL, median [IQR]	2096 [1167–4345]	4891 [2455–8230]	0.001
D-dimer, µg/mL, median [IQR]	4.845 [2.670–10.96]	8.365 [3.325–18.29]	0.083
Prothrombin fragment 1+2, pg/mL, median [IQR]	4377 [3005–5433]	3724 [2236–5964]	0.370

Continuous variables are shown as median (IQR) and compared using the Wilcoxon rank-sum or Student's t-test; categorical variables are shown as count (%) and compared using Fisher's exact or Pearson's Chi-square test,

as appropriate. AKI is defined according to the RIFLE criteria. Compared to patients without AKI, those with AKI exhibited a higher Injury Severity Score (ISS), greater incidence of traumatic brain injury (TBI), larger transfusion volumes, higher creatinine on arrival, lower eGFR, and more pronounced hemostatic abnormalities. Statistically significant tests ($p < 0.05$) are highlighted in bold.

Abbreviations: ISS, Injury Severity Score; TBI, traumatic brain injury; SBP, systolic blood pressure; HR, heart rate; GCS, Glasgow Coma Score; TXA, tranexamic acid ; RBC, red blood cells ; eGFR, estimated glomerular filtration rate; Hb, hemoglobin ; INR, international normalized ratio.

Table 2. Composite biomarker model results.

	Univariate			Multivariate		
	OR	95% CI	p	OR	95% CI	p
Age	1.03	0.999 – 1.055	0.057	1.03	0.988 – 1.077	0.164
Male sex	4.83	0.617 – 37.87	0.134			
BMI	1.01	0.959 – 1.073	0.610			
Race	0.64	0.191 – 2.115	0.460			
ISS	1.06	1.027 – 1.100	< 0.001	1.05	0.999 – 1.114	0.054
Mechanism of injury	0.53	0.214 – 1.312	0.169			
Confirmed TBI	4.56	1.227 – 16.95	0.024			
Arrival SCr	18.1	3.679 – 88.90	< 0.001	54.1	5.497 – 533.1	0.001
Arrival Hb	0.97	0.786 – 1.190	0.754			
Arrival Lactate	1.06	0.942 – 1.198	0.322			
Arrival INR	0.83	0.519 – 1.319	0.426			
Arrival Platelet count	0.99	0.986 – 0.998	0.011			
Shock index	2.32	0.943 – 5.700	0.067			
Total weight-adjusted transfusion volume in 24h	1.01	1.003 – 1.012	< 0.001			
Given TXA dose	1.00	0.783 – 1.277	1.000			
Six-hour TXA concentration	1.04	1.013 – 1.073	0.005			
Six-hour PAP-to-fibrinogen ratio	1.01	1.000 – 1.029	0.048			
Six-hour TAT-to-fibrinogen ratio	5.82	2.642 – 12.80	< 0.001	7.17	2.344 – 21.91	0.001

The composite biomarker model assessed the association between 6-hour ratios and the development of acute kidney injury (AKI), defined by the RIFLE criteria. Age and Injury Severity Score (ISS) were forced into all models. Odds ratios (OR), 95% confidence intervals (CI), and p values are shown for both univariate and multivariable analyses. In the multivariable model, a higher TAT-to-fibrinogen ratio was independently associated with increased odds of AKI. The model included 116 patients (92 without AKI, 24 with AKI) and demonstrated good explanatory power (Pseudo- $R^2 = 0.42$; LR $\chi^2(4) = 49.4$; $p < 0.001$) and excellent discrimination (AUC = 0.88). Model calibration was assessed using the Hosmer-Lemeshow goodness-of-fit test ($\chi^2(8) = 7.25$, $p = 0.510$), indicating no evidence of lack of fit. Statistically significant associations ($p < 0.05$) are highlighted in bold.

Abbreviations: ISS, injury severity score ; TBI, traumatic brain injury ; Hb, hemoglobin ; INR, international normalized ratio ; OR, odds ratio ; CI, confidence interval ; TXA, tranexamic acid ; LR : likelihood ratio ; AUC, area under the curve.

Table 3. Predictive efficiency of TAT, PAP, and fibrinogen in adult trauma patients with acute kidney injury.

	AUC	95% CI	Cut-off value	Sensitivity	Specificity	PPV	NPV
Arrival TAT	0.667	0.568–0.767	≥ 116 pg/mL	85.7%	49.5%	32.4%	92.5%
Six-hour TAT	0.819	0.730–0.909	≥ 99 pg/mL	85.2%	67.4%	43.4%	93.9%
Arrival fibrinogen	0.653	0.540–0.765	≤ 171 mg/dL	70.4%	63.7%	32.7%	87.8%
Six-hour fibrinogen	0.664	0.541–0.786	≤ 173 mg/dL	57.7%	81.2%	44.1%	88.2%
Arrival PAP	0.634	0.521–0.746	≥ 2190 pg/mL	76.9%	51.1%	31.7%	88.2%
Six-hour PAP	0.705	0.598–0.813	≥ 4302 pg/mL	65.4%	73.0%	41.5%	87.8%

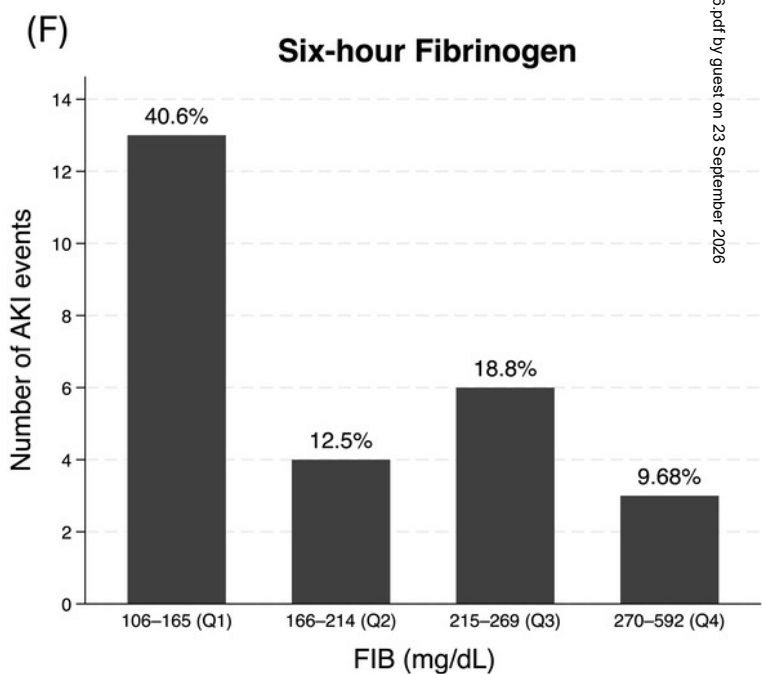
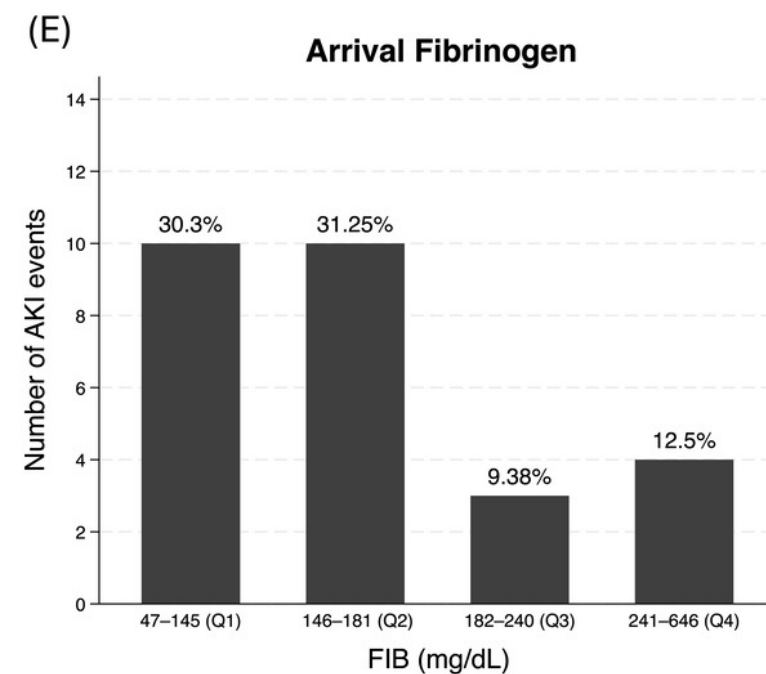
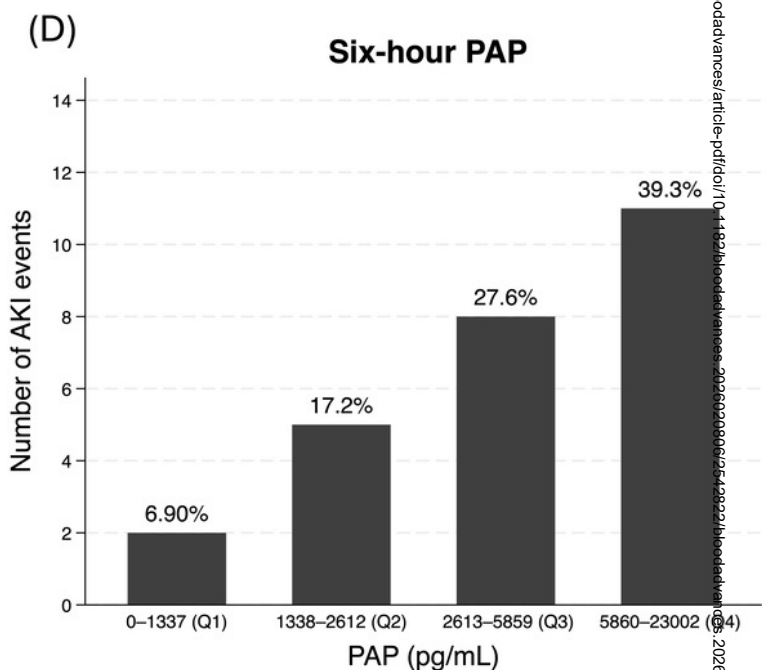
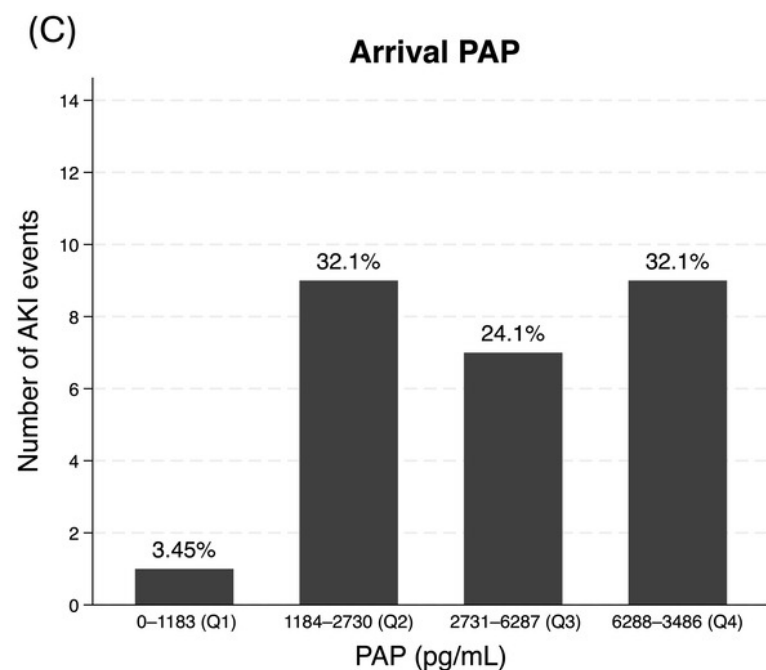
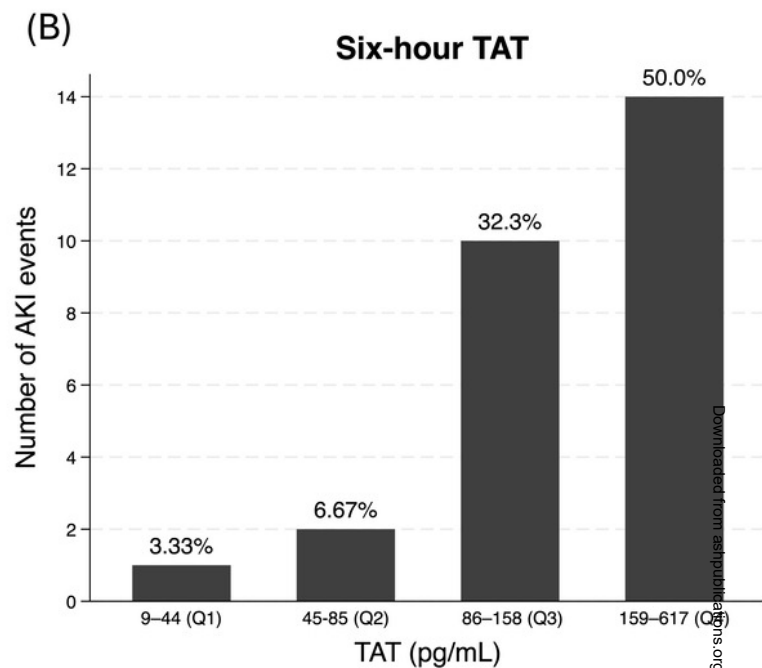
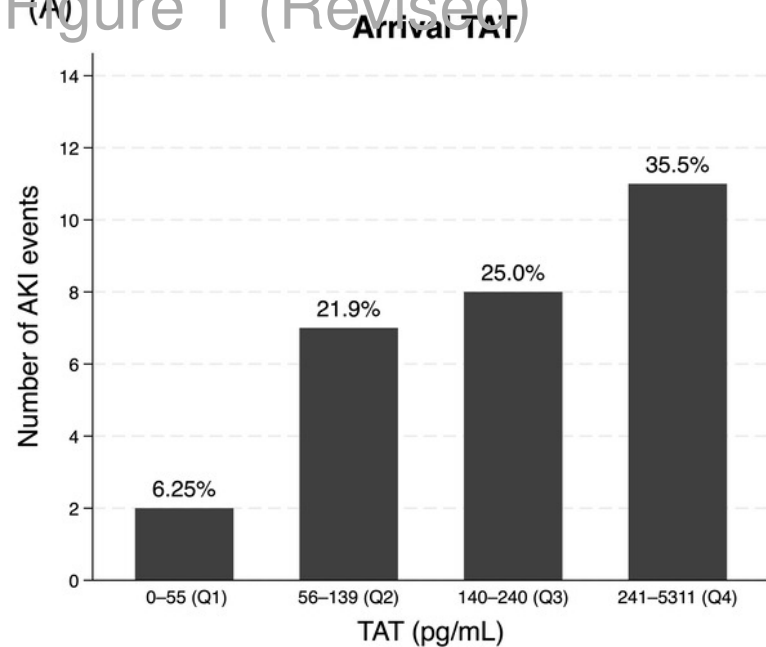
Diagnostic performance metrics are shown for arrival and 6-hour levels of thrombin-antithrombin (TAT) complex, plasmin- α 2-antiplasmin (PAP), and fibrinogen in predicting AKI (RIFLE criteria). 6-hour TAT demonstrated the strongest discrimination (AUC = 0.819) with high sensitivity (85.2%) and specificity (67.4%) at cut-off of ≥ 99 pg/mL. Fibrinogen and PAP had moderate predictive value, with 6-hour fibrinogen showing the highest specificity (81.2%). Cut-off values were derived using the highest Youden Index. Abbreviations: AUC, area under the curve ; CI, confidence interval ; PPV, positive predictive value ; NPV, negative predictive value ; TAT, thrombin-antithrombin complex ; PAP, plasmin- α 2-antiplasmin complex.

Table 4. Clinical Outcomes stratified by high and low Thrombin-Antithrombin (TAT), Plasmin-Antiplasmin (PAP) and fibrinogen levels using derived thresholds.

	Low TAT (N = 66)	High TAT (N = 53)	Low PAP (N = 74)	High PAP (N = 41)	Low Fibrinogen (N = 34)	High Fibrinogen (N = 93)
AKI, n (%)	4.0 (6.1%)	23 (43%) ^{***}	9.0 (12%)	17 (42%) ^{***}	15 (44%)	11 (12%) ^{***}
MOF, n (%)	11 (17%)	27 (51%) ^{***}	16 (22%)	21 (51%) ^{**}	16 (47%)	21 (23%) ^{**}
ARDS, n (%)	1.0 (1.5%)	9.0 (17%) ^{**}	3.0 (4.1%)	7.0 (17%) [*]	4.0 (12%)	6.0 (6.5%) ^{ns}
Mortality, n (%)	1.0 (1.5%)	4.0 (7.6%) ^{ns}	1.0 (1.4%)	4.0 (9.8%) ^{ns}	3.0 (8.8%)	2.0 (2.2%) ^{ns}
Length of stay, days, median [IQR]	8.0 (5.0–11)	14 (8.5–21) ^{***}	8.5 (5.0–13)	13 (8.5–22) ^{**}	13 (7.0–21)	9 (5–14) ^{ns}
ICU admission, n (%)	52 (79%)	48 (91%) ^{ns}	58 (78%)	38 (93%) ^{ns}	30 (88%)	72 (77%) ^{ns}
ICU-free days, median [IQR]	28 (27–30)	26 (17–28) ^{***}	28 (26–30)	26 (14–28) ^{***}	27 (17–28)	28 (26–30) [*]
Mechanical ventilation, n (%)	33 (50%)	40 (76%) ^{**}	35 (47%)	35 (85%) ^{***}	24 (71%)	51 (55%) ^{ns}
Ventilator-free days, median [IQR]	30 (28–30)	28 (23–30) ^{***}	30 (28–30)	28 (22–29) ^{***}	29 (25–30)	30 (28–30) ^{ns}

Patients were grouped by biomarker levels based on optimal cut-off values identified using the highest Youden Index from receiver operating characteristic (ROC) analysis: TAT ≥ 99 pg/mL, PAP ≥ 4302 pg/mL, and fibrinogen ≤ 173 mg/dL. High TAT and PAP levels corresponded with increased incidence of acute kidney injury (AKI), multiple organ failure (MOF), and indicators of more severe clinical trajectories, including longer hospital and ICU stays. Low fibrinogen levels also reflected higher rates of AKI and MOF, though differences in other outcomes were less distinct. Categorical variables were evaluated using Fisher's exact or Pearson's Chi-square test; continuous variables using the Wilcoxon rank-sum or Student's t-test, as appropriate. AKI was defined by the RIFLE criteria, MOF by the Marshall score, and Acute Respiratory Distress Syndrome (ARDS) by the Berlin definition. ns not significant, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Figure 1 (Revised)



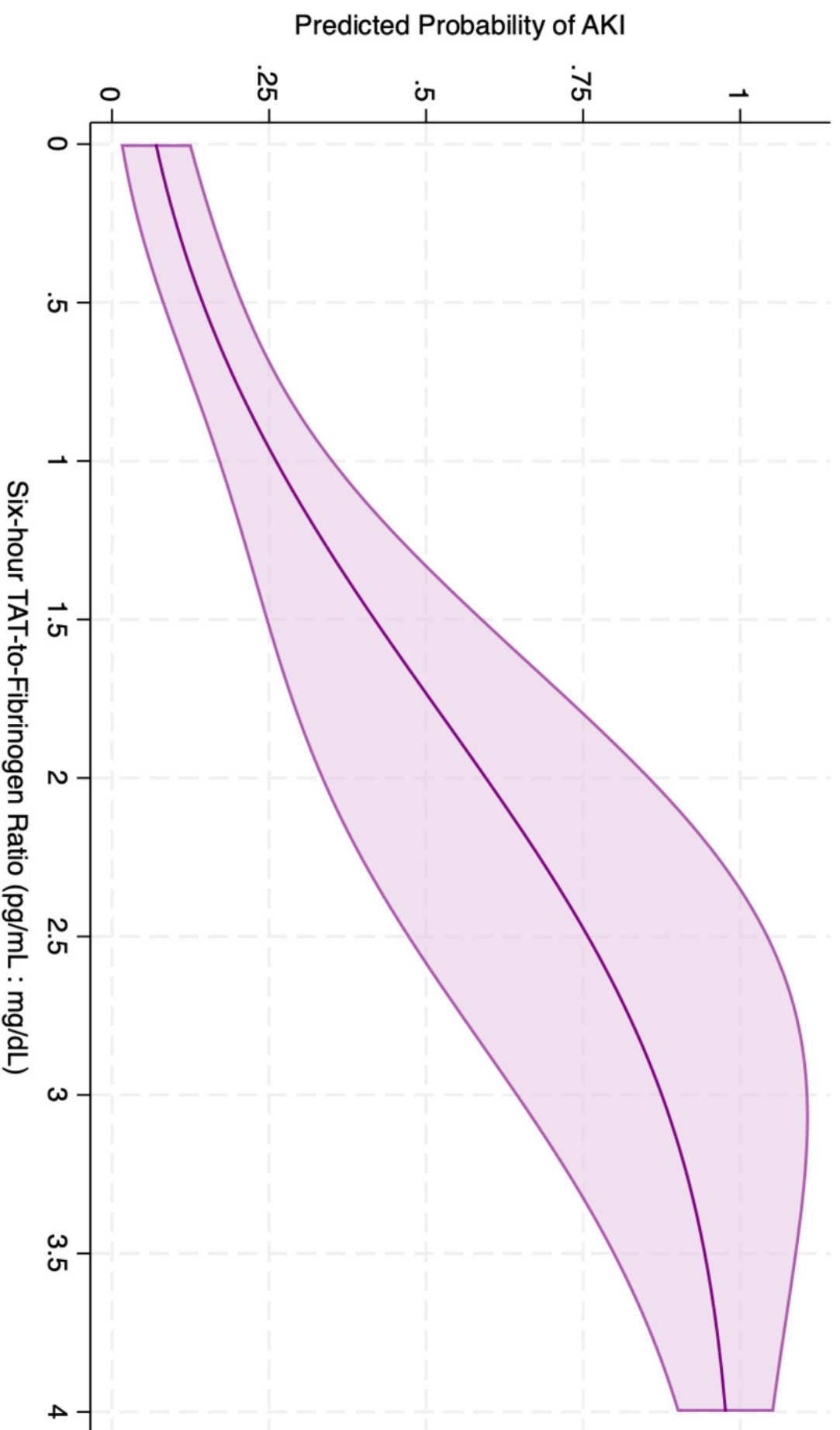


Figure 3 (Revised)

